

Long draft

Motivation as engineer: beat limitations using intelligence, to change the world using my hands and mind, leading others showing them the path to discover new technologies.

The crowd roaring, you can feel the tremble inside your chest. You listen to the cheering from your team mixed with the noise across the field. Final check all the systems and you step out of the field. Volunteers close the gates and you watch across the other alliance section. World champions with machines 3 or 4 times your team budget, top engineering and more years of experience. Looking back, and there they are, all your students ready to watch the robot move. In that moment, you feel the weight of nights without sleep, countless hours of building, wiring and testing, all for that moment. So, when your machine crosses the field at the top speed using the paths and routines that you programmed, the mechanisms that you wired, all together surpassing the top robots of the world, it's impossible to hold the tears.

I'm Cristopher Hernández and this is the story on why I became an engineer. After 3 and a half years as FRC member, I learned that as engineers we can shape the world with our hands and minds, regardless of the budget or limitations we have.

But that's something I learned as a child. Thanks to my job we had to move to Matehuala, a small town of the state of San Luis Potosí. I did so well on math and physics that one of my teachers invited me to participate on a science fair and we won. We had the opportunity to present at Expo Science International ESI 2019 in Abu Dhabi and the CIENTEC 2019 at Peru. The question was not how we get there is what we are presenting, because my father started working double to gather the money to travel, we sold popsicles, throwed parties and did shuffles to achieve the goal.

At those events I discover lots of projects from all over the world, so when I came back my idea was clear, work on science and technology. That time, the pandemics got active at Mexico and we had to go home. Since I bought my first computer to participate in the events, it was me and the computer alone during two years. The time came when I played all the maps on the different videogames and got bored. Instead of waiting, I decided to learn how to code and build my own maps, and then my uncle gave me a kit of Arduino UNO and my passion grew. Imagine a child with 10 hours of programming and experimenting with electronics.

Moved by my passion, I researched for the best high schools on the zone and I found the Tecnológico de Monterrey. It was super expensive, but I worked hard to got high score on the admission exam and obtain a scholarship. The objective? Get into the best robotics team at Mexico, team 3478 which was the perfect fit to depart my career. Since learning

mechanics online was hard, I entered directly to programming and electrics area. There I met awesome mentors driven by the love of engineering and FRC games.

I started as a rookie, that first year I became robot main technician, in charge of validating the final settings at the fields. Second year as senior technician and my third year as former mentor of the area. We won several regional competitions and arrived at the world championships division final three years in a row. We also got the autonomous engineering awards sponsored by Ford and GM.

The final semester when I graduated from I won the Life Medal, maximum galardón for working on extracurricular activities, including the world championship on FRC, funding a math club and participation on the school musical Mamma Mia on the music ensemble as second pianist. I also presented a TedTalk on one of the first editions of TedX TecDeMonterrey SLP, bringing the spot to my passion also on public speaking and as final act, I was in charge of graduation speech in front of all the generation which was a tremendous moment.

After that, I had already decided to study mechatronics engineering. My path was to accept a job as former mentor of the team 3478 as programming and electric mentor. I stayed there from 6 months working in parallel at the VR zone of the campus library to expand and take care of it. I learned management and about edge technology, then the scholarship was not enough to keep up and I had to drop, but instead of staying stale, I decided to move up.

That past summer, before entering Tec college, I had the chance to participate in a robotics camp teaching also C++ and Python with children and teens. I received a recommendation from a friend to apply to AI training jobs, so I got one spot at Scale AI as AI trainer, testing some model responses, paths to get the model to work better with user actions.

After that experience I presented the admission test to the engineering school at UASLP, where I got the first place. Once in, I dedicated my time to study, get my classes ready and also participate in Magna Hackathon 2024 with honorific mention on a system with machine learning to predict employee rotations on enterprises based on employee data, and the 2024 ICP competition getting top 3 of the school place. After that, early 2025 I got a job as 3D printing engineer at a new startup on my region, LEAD3D printing, I worked with CAD, Slicers and Bambu lab printers to adjust different materials, temperatures, structures and optimize the business jobs and inventories. I also got the chance to use Raptor scanner and reconstruction technology to perform reverse engineering to pieces and even chairs.

That was the moment the new opportunities arrived, I participated in the program Technovation Girls mentoring a team for a full region and also giving a talk about

programming and algorithms. We developed an app for women to get close to jobs, training and community.

I got works on personal AI and programming projects, until a friend talked about an option at Oracle. I went for it, after failing at Meta, and other start ups this was a new opportunity at my door. I just sent my CV application and one day I received response back from recruiters. The message was clear, my profile was standing out and they wanted a first interview.

At that, I presented my first round and then a second round with my manager and it was the best, I got the job and immediately I got to start working on that. I accepted and went to a training at Zapopan Guadalajara, where the Mexican Development Center from Oracle, I met incredible people and had the chance to understand the reach of AI and software tech on first glance. The same time, brand new protocols touched the software section and my job was to get them working and show as demos, MCP, A2A, Workflows, GenAI from Oracle Cloud, LangChain, LangGraph and more, but most proud about the integration of observability and tracing for AI flows using LangFuse, a framework that could be self-hosted, so I did one of the first efforts to use this inside the org. Today is wide used across different teams and as a former solution on different products.

I also worked with a wide team to integrate AG-UI protocol released some weeks early after some demos, we gather a team to develop a PoC with customers to inject real energy data and display on the screen as graphs, visuals integrated and personalized for the user, we worked on the agentic layer and I was in charge of two different modules. Then, I worked on a team to show a toolkit and developed a solution using GraphDB and Oracle DB for a full three app pipeline that used an agent to orchestrate the flow and present visual data, but another one that was a textual fast agent using tools to fetch data in order to present results to the user, using the A2UI latest protocol derived from AG-UI just some weeks ago.

That result was presented in a customer conference as result from our department. Right now I'm working on OpenAI SDK to use GenAI services and OpenAI models to build flows, automations and code executions with skills.

The langfuse integration was used and I presented it on the OCI GenAI workshop at Guadalajara MDC center this past April 2026, I presented and shared with the teams. The LangGraph and LangChain codes and modules were presented on the India 2025 November workshop OCI GenAI at India. During this time I also had a touch point on mock interviews as volunteers with Forge foundation, where I helped them polish their paths and abilities to apply for jobs.

After I got my job at Oracle, I left French classes because I thought that I would not have the time, it was a moment when I thought the dream was over, but after that first semester, my manager granted me the extension, and I decided that I can keep up, so I signed for French classes until today. I decided that the shot at the scholarship was the best option, because at Oracle I learned the basics, but I need the boost to open the door behind me.

When I had my first semester at Tec college, I was aiming for DAAD scholarship, which allowed me a semester at Germany, but then I left and I found a better option at UASLP to go to France, which allowed me to not only complete a semester, but a double degree at France and get me into the European enterprises.

This is the past, how I built the ideals on why robotics, and I know that I didn't explained because I was into software, but I always felt that I'm missing a piece of myself, seeing me as stuck on the computer but not a real impact on my daily life. So I took all the chances and the France environment is the perfect fit to get into Europe, develop and research of robotics there are the required tools and specially for the schools at the Central program.

They have labs that can get me into the research of collaboration of robotics, I believe that the next revolution is on the Industry 4.0, not AI on computers, rather how they integrate in every day tasks, right now all is abstract and we talk about workflows and doing things in seconds, but not really, we still have to do laundry, wash the dishes, drive cars and work at factories or doing process that are non linear or deterministic. I have a strong foundation on how the AI systems work and how they are shaping the digital work, but I want to take the next step and bring that advantage into the real world to solve real problems with hardware and software, which is exactly what a mechatronics engineering does.

The central schools at France have advanced robotics programs, research and development in their industries, Startups that are already working on cobots and detection, vision systems, and I can get the foundations on how to integrate the latest AI systems into the physical world, with the cutting edge technology along with the safety of the deterministic machines. Right now, the trust on the AI integration on big industries, energy, manufacturing, assembly, all rely only on the tested safe methods, because AI is so unpredictable that the flows have to be controlled, limited to what we can see. In the future we will need engineers that understand both deterministic and intelligent systems and how to integrate them with safety, to increase productivity keeping reliability.

My final goals are to get the best of AI and machine learning models into the industry to close the gaps on small and medium industries sections with the large scale enterprise, because, big companies can work with a multimillion dollar robotic cells and armies of arms and robots, but a small business maybe will only be able to buy one or two cells, so

we need the flexibility to make that work on different environments and on different tasks with the same hardware.

The Bajío region at Mexico has a need of technological innovation that can be fed directly from France and integrate. France has the need to integrate strong and reliable technologies into lines, which can be achieved by experience and combination of work. This is the future of engineering and we need to work on it.

But what if we take this further, like, apply ML and AI into energy distribution systems, batteries, homes, make this reliable and smart, that is currently implemented, and is how the future systems must work. There is a motto at First, which says “safety first” which means, before moving, always check safety and reliability, so that applied to energy to keep business, hospitals, transport, cities, is required. The experience on the moving systems along with the energy usage is what we are doing and working towards.

As an engineer, I want to close that gap, have reliable systems that can be controlled but also smart. Systems that work on the hard conditions on development countries and the most technologically advanced ones. I want to build bridges that allow people to surprise and walk to the edge technologies. As engineers, we are curious and hard working, we surpass the limitations of life, just as I did before, I want to offer that option to others, work on quality and excellence.

France is the best option and ecole centrales are the best program to achieve this, prepare a young person on AI, robotics and Ambiental scopes that we miss on Mexico, educational programs are different, for UASLP focused still on learning the engineering, but at France and Europe you are worried about Ambiental, consumption, responsible usage, renewal which is years ahead of Mexico, so imagine if we implemented smart systems that are on early stages at France into the Mexico ecosystem, or any other country that has received heavy inversion of industries, we would reduce the Ambiental impact, increase reliability and show people on those regions what is possible at top technologies, that's the bridge I want to build.

I want to work on smart systems, reliable enough to have flexibility, safe and of course, reducing the impact on the Ambiental agenda. I also want to open this world to others using my communication skills, approach to people and their environment to understand the needs and apply real solutions.